

Syllabus - Math 19620, Section 20

Instructor: Miles Kretschmer

Spring Quarter 2026

This syllabus may be updated during the quarter, the most recent version will be on the Canvas page.

1 Information for Reference

- Instructor Information:
 - Email: mjkretschmer@uchicago.edu
This is the best way to reach me. I am happy to respond to questions about the course over email.
 - Office: Eckhart 7, in the basement of Eckhart Hall
- Lectures: Tuesdays and Thursdays, 9:30 AM - 10:50 AM, in Eckhart 203
- Office Hours: by default, these will be held in my office.
 - Monday: 3 - 4 PM
 - Friday: 4:30 - 6:30 PM

If these times do not work for you, feel free to email me to schedule a meeting at a different time.

- Textbook: Bretscher's *Linear Algebra with Applications*, 5th edition. A pdf can be found on Canvas.
- Homework: A problem set will be assigned on Wednesday or Thursday each week except 9th week, and will be due 10:00 PM on Thursday the next week. There will be 8 problem sets in total.
For policies regarding homework, see section 4.
- Exams: There will be one midterm and a final. The midterm will be on Thursday of week 5, at lecture time: 9:30 - 10:50. The final will take place on Friday, May 29th from 1:00 - 3:00.
For policies regarding exams, see sections 5 and 6.1.
- Grading: 40% of the grade is homework, the midterm is 25% and the final 35%.

2 Course Description and Objectives

This course is an introduction to linear algebra. This begins with the study of systems of linear equations. From there, we will introduce the algebra of vectors and matrices, linear transformations, subspaces, and dimension. After this, we will study length, angle and orthogonality through the inner product, including orthogonal projection onto subspaces and its application to least squares approximation. We will then introduce determinants and the characteristic polynomial, and use this to find eigenvalues of matrices and their eigenvectors. We will conclude by studying diagonalization of matrices and its application to discrete time dynamical systems. Throughout the course we will consider applications of these tools in natural and social science. By the end of the quarter, you will be able to:

1. Construct a valid and precise mathematical argument.
2. Determine when a system of linear equations has a solution, and describe the set of solutions.
3. Represent systems of linear equations using matrices, and carry out algebraic computations with matrices including matrix multiplication and inversion. Describe in geometric terms the linear transformation represented by a matrix.
4. Describe linear subspaces of $\mathbb{R}^n$ in terms of defining equations, spanning sets and bases, and determine their dimensions. Determine the image and kernel of a linear transformation.
5. Find orthonormal bases using the Gram-Schmidt process. Find the orthogonal complement of a subspace and the orthogonal projection of a vector onto a subspace. Apply orthogonal projection to least squares approximation problems.
6. Compute the determinant of a matrix, and use this to determine if it is invertible and draw geometric conclusions about the linear transformation it represents.
7. Find (real and complex) eigenvalues and eigenvectors of a matrix. Determine if a matrix can be diagonalized, and find a diagonalization. Use eigenvalues and diagonalization to describe the long term behavior of discrete time dynamical systems.

3 Planned Schedule

This is a plan and plans change. The numbers refer to sections of the textbook in which the topic is discussed.

Week 1 3/24 & 3/26	Systems of linear equations. (1.1,1.3) Introduction to vectors in $\mathbb{R}^n$, linear combinations. (1.2) Matrix representations of linear systems. (1.3)	
Week 2 3/31 & 4/2	Matrices and linear transformations. (1.3,2.1) Geometry of linear transformations. (2.2) Matrix multiplication. (2.3) The inverse of a matrix. (2.4)	4/2 Problem set 1 due
Week 3 4/7 & 4/9	More on Inverse matrices. Linear subspaces of $\mathbb{R}^n$, Span. (2.4, 3.1, 3.2) The image and kernel of a linear transformation. (3.1)	4/9 Problem set 2 due
Week 4 4/14 & 4/16	More on image and kernel. (3.1) Linear dependence and bases. (3.2) Dimension, Bases as coordinate systems. (3.3, 3.4)	4/16 Problem set 3 due
Week 5 4/21 & 4/23	Length, angle, and orthogonality. Orthonormal bases. Orthogonal projection onto a subspace. (5.1)	4/23 Midterm, Problem set 4 due
Week 6 4/28 & 4/30	The Gram-Schmidt process and orthogonal projection. (5.1, 5.2) Least squares approximation. (5.4)	4/30 Problem set 5 due
Week 7 5/5 & 5/7	Determinants. (6.1,6.2) The characteristic polynomial, eigenvalues and eigenvectors. (7.1,7.2)	5/7 Problem set 6 due
Week 8 5/12 & 5/14	More on eigenvalues and eigenvectors. Diagonalization of matrices and its uses. (7.1,7.2,7.3)	5/14 Problem set 7 due
Week 9 5/19 & 5/21	Complex eigenvalues and eigenvectors. (7.5) Discrete dynamical systems and linear recurrences. (7.4)	5/21 Problem set 8 due

4 Homework Policies

On each problem set some problems will be graded for correctness, and the rest for completion. The purpose of the problem sets is to give you practice with the things introduced in lecture, and working through them

will be an important part of learning the material of the course. Because of this, I ask you to not use AI when working on them.

4.1 Late Policy

You can submit up to two problems sets up to two days late, which comes out to 10:00 PM on Saturday. If you will submit a problem set late, email me to let me know of this by the Thursday that the problem set is due. Outside of this, late work will not be accepted.

4.2 Collaboration

Discussing the problem sets with other students is encouraged. When working with others, you should write up your own solutions in your own words, and make sure that you can carry out computations on your own. I also ask that you indicate on the problem set the people with whom you worked.

5 Exam Policies

Problems on the exams will be similar to homework problems, so the best way to prepare for the exams is to work through the problem sets. In week 9, I will send out a set of review problems. This will be formatted like a problem set, but will not be collected or graded.

The exams will be closed book, with no electronic devices. Solutions should come with justification for full credit. Partial credit will be given for partial solutions. Please let me know as soon as possible if some exceptional circumstance prevents you from taking the exam. In normal circumstances, there will be no make up exams.

6 University Policies

Add/drop: Add/drop for all courses ends Friday Week 3, on April 3. This is the last day you can drop a class without it being present on your academic transcript.

Withdraw (W): After the add/drop deadline, you can still withdraw from the course up to Monday, May 18 in Week 9 at 5pm, but it will result in a W (withdraw) grade on your transcript.

Incomplete (I): If emergency circumstances prevent you from taking the final exam, you may be eligible to request an “I” (incomplete). A request form must be signed by your academic adviser and your instructor.

Pass/fail: You can request pass/fail option (opting out of a letter grade) by emailing me up until Friday, May 22 in Week 9 at 5pm.

6.1 Final Exam

It is the policy of the Department of Mathematics that the following rules apply to final exams in all undergraduate mathematics courses:

The final exam must occur at the time and place designated on the College Final Exam Schedule. In particular, no final examinations may be given during the Ninth Week of the quarter. Instructors are not permitted to excuse students from the scheduled time of the final exam except in the cases of an Incomplete.

6.2 Disability Accommodations

Students with disabilities who have been approved for the use of academic accommodations by Student Disability Services (SDS) and need a reasonable accommodation(s) to participate fully in this course should follow the procedures established by SDS for using accommodations. Timely notifications are required in order to ensure that your accommodations can be implemented. Please meet with me to discuss your access needs in this class after you have completed the SDS procedures for requesting accommodations. SDS can

be reached at:
Phone: (773) 702-6000
Email: disabilities@uchicago.edu